Colorectal Polyp Segmentation: Comparing CNN and Transformer Models and Improving Out-of-Domain Generalization

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Abstract

Purpose: Polyp segmentation models reach a Dice score close to 0.90 on images from the centers they were trained on, but lose much of it on images from other centers: PraNet scores 0.898 on Kvasir-SEG and 0.628 on ETIS. Most published comparisons also rely on Dice alone. This study compares CNN and Transformer segmentation models under one training pipeline and tests whether low-cost changes narrow the out-of-domain gap.
Methods: All models are trained on the PraNet split of Kvasir-SEG and CVC-ClinicDB (1,450 images) and tested on five sets, three of which come from centers unseen during training. We reproduce U-Net, MultiResUNet, CE-Net, and TransUNet [and Swin-Unet] with the same data, augmentation, loss, and schedule, then add a boundary-aware loss and colour-based domain augmentation. Models are compared with Dice, IoU, recall, and HD95, and differences are tested with the Wilcoxon signed-rank test on per-image scores.
Results: [Placeholder] Dice of each model on the in-domain and out-of-domain sets; size of the generalization gap; whether Transformers beat CNNs significantly; effect of each improvement in the ablation; inference speed.
Conclusion: [Placeholder] One or two sentences on the main finding, followed by the main limitation (small training set, public benchmarks only, few runs per configuration).

Keywords: polyp segmentation, colonoscopy, U-Net, Vision Transformer, domain generalization, boundary loss, statistical testing

1 Introduction

Most colorectal cancers develop from polyps in the colon, and finding and removing polyps during colonoscopy lowers the risk of cancer [citation]. Polyps vary in size, shape, and colour, and many look similar to the surrounding mucosa, so some are missed. Automatic segmentation outlines the polyp pixel by pixel and can support the endoscopist during the procedure. Encoder–decoder networks derived from U-Net [1] are the standard approach, and recent models extend or replace the convolutional encoder with Transformers [2, 3].

Scores on the usual benchmarks are high, but they drop sharply on data from other centers. PraNet reaches a mean Dice of 0.898 on Kvasir-SEG and 0.899 on CVC-ClinicDB, the two sources of its training data, but 0.709 on CVC-ColonDB and 0.628 on ETIS [4]. Evaluation practice adds a second problem. An audit of 27 polyp segmentation papers found that 26 report no significance test, 25 omit HD95, and 25 omit recall; on the six unseen centers of PolypGen, all re-evaluated models fell to a Dice of 0.71–0.76 [5]. Small Dice differences measured on in-domain sets therefore say little about which model works better on new data.

This report does not attempt to set a new state of the art. It studies a narrower question: under one controlled pipeline, how do the CNN and Transformer models of the course reading list [1, 2, 3, 6, 7, 8] compare on in-domain and out-of-domain data, and can low-cost changes reduce the gap between the two?

The report addresses three research questions:

- **RQ1.** With about 1,450 training images, do Transformer-based models (TransUNet, Swin-Unet) outperform CNN-based models (MultiResUNet, CE-Net) by a statistically significant margin?
- **RQ2.** Which model keeps its performance best on the out-of-domain sets CVC-ColonDB, ETIS, and CVC-300?
- **RQ3.** How much do a boundary-aware loss and colour-based domain augmentation reduce the generalization gap, and which component contributes most?

The report is organized as follows. Section 2 reviews encoder–decoder CNNs, Transformer-based segmentation, and polyp benchmarks. Section 3 defines the task and the data. Section 4 describes the experimental pipeline and the models, followed by the loss functions and the proposed improvements in Section 5. Sections 6 and 7 present the experimental setup and results. Section 8 states the reliability constraints, and Section 9 concludes the study.

2 Related Works

2.1 Encoder–decoder CNNs for medical image segmentation

U-Net combines a contracting encoder with an expanding decoder and passes high-resolution features through skip connections, which lets it learn from few annotated images [1]. MultiResUNet replaces the two convolutions of each stage with a MultiRes block, a chain of 3×3 convolutions whose outputs are concatenated to cover several receptive fields, and adds residual "Res paths" to the skip connections to reduce the

semantic gap between encoder and decoder features; it was evaluated on five biomedical datasets, including CVC-ClinicDB and ISIC 2018 [6]. CE-Net places a dense atrous convolution (DAC) block and a residual multi-kernel pooling (RMP) block between a pretrained ResNet encoder and the decoder, so that multi-scale context is captured without further downsampling [7]. nnU-Net proposes no new architecture: it configures preprocessing, network topology, training, and post-processing from a "fingerprint" of the dataset and was evaluated on the ten tasks of the Medical Segmentation Decathlon [8].

2.2 Transformers for medical image segmentation

Convolutions capture local patterns, and their limited receptive field makes long-range context hard to model. TransUNet passes CNN feature maps to a Vision Transformer encoder and decodes them with a cascaded upsampler and U-Net skip connections [2]. Swin-Unet builds the whole U-shaped network from Swin Transformer blocks, whose shifted windows reduce the cost of self-attention, and upsamples with patch-expanding layers [3]. Both models were evaluated on the Synapse multi-organ CT and ACDC cardiac MRI datasets rather than on colonoscopy images.

2.3 Polyp segmentation benchmarks and evaluation practice

PraNet introduced the protocol that most polyp papers follow: training on 900 Kvasir-SEG and 550 CVC-ClinicDB images and testing on five sets [4]. PolypGen adds images from six centers to test generalization [9]. The audit in [5] found at least five incompatible split protocols in use, transcription errors in survey tables, and differences of up to about 16 px in HD95 between metric libraries. It proposes a reporting checklist: HD95 and recall as primary metrics, an explicit split protocol, an external test set, and significance tests over per-image scores. Metrics Reloaded gives general guidance on choosing segmentation metrics [10]. This report follows these recommendations.

3 Problem Formulation and Data

3.1 Task formulation

An image $x \in \mathbb{R}^{H\times W\times 3}$ has a binary mask $y \in \{0,1\}^{H\times W}$ that marks the polyp pixels. A model f_θ outputs a probability map, and the predicted mask is obtained by thresholding:

$$p = f_\theta(x) \in [0,1]^{H\times W}, \qquad \hat{y} = \mathbb{1}[\,p \geq 0.5\,]. \tag{1}$$

Training images come from a source domain $\mathcal{D}_S$ (Kvasir-SEG and CVC-ClinicDB). Test images come from $\mathcal{D}_S$ and from K target domains $\mathcal{D}_T^1, \ldots, \mathcal{D}_T^K$ that are never seen during training (CVC-ColonDB, ETIS, and CVC-300). For a per-image score s (Dice in the main analysis), the generalization gap of a model is

$$\Delta = \bar{s}\big(\mathcal{D}_S^{\text{test}}\big) - \frac{1}{K}\sum_{k=1}^{K} \bar{s}\big(\mathcal{D}_T^k\big), \tag{2}$$

Table 1: Datasets and their roles.

Dataset	Images	Resolution	Role
Kvasir-SEG [11]	1,000	varied	Training; in-domain test
CVC-ClinicDB [12]	612	384×288	Training; in-domain test
CVC-ColonDB [13]	380	574×500	Out-of-domain test
ETIS-LaribPolypDB [14]	196	1225×966	Out-of-domain test
CVC-300 (EndoScene) [15]	60	[verify]	Out-of-domain test

Table 2: Number of images per split.

Split	Kvasir	ClinicDB	ColonDB	ETIS	CVC-300	Total
Training	810	495	–	–	–	1,305
Validation	90	55	–	–	–	145
Test	100	62	380	196	60	798

where $\bar{s}$ is the mean score over the images of a set. A model generalizes better when it scores high on the target domains and has a small Δ.

3.2 Data sources

Table 1 lists the five public datasets. Kvasir-SEG and CVC-ClinicDB provide the training images; CVC-ColonDB, ETIS, and CVC-300 are used only for testing. All five are used in the form released with PraNet [4].

3.3 Data split

Table 2 gives the number of images per split. Training and test images follow the fixed PraNet split [4]. Ten percent of the training images of each source dataset are held out at random (seed 42) for validation, so that checkpoints are selected without touching any test set. Because the three CVC datasets were collected by related groups, near-duplicate images across the training and test sets are checked with perceptual hashing before training, as in [5]. Figure 1 shows sample images.

4 Experimental Pipeline

Figure 2 shows the pipeline, and Table 3 lists the models. Every model goes through the same stages with the same data, augmentation, loss, and schedule; only the architecture changes, so differences in the results can be attributed to the architecture rather than to the training recipe. The shaded stages are where the improvements of Section 5 are applied.

TransUNet and Swin-Unet expect 224×224 inputs by default, while the CNNs follow PraNet and use 352×352. The input size of each model is reported in Appendix A, and predictions are resized to the original resolution before evaluation, so

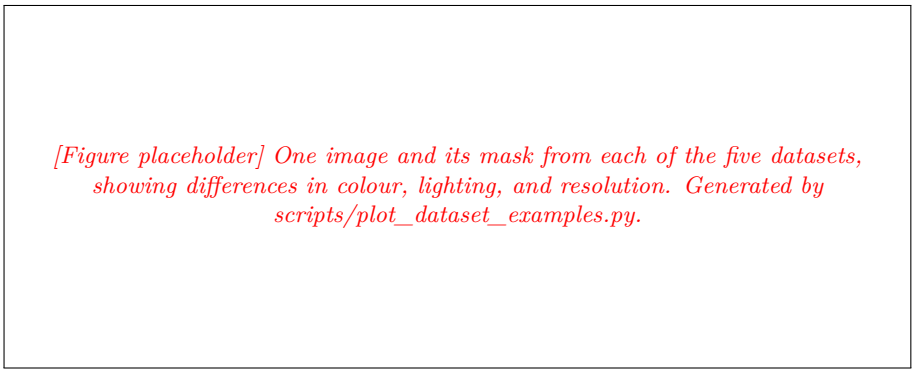


Fig. 1: Sample images and ground-truth masks from each dataset.

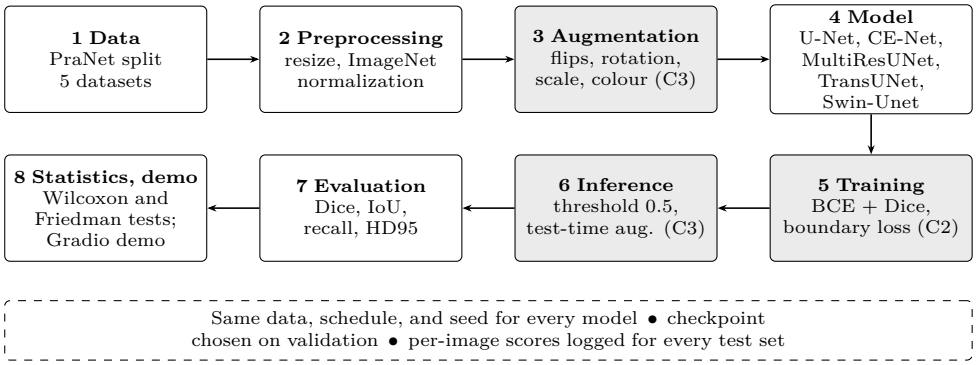


Fig. 2: Experimental pipeline. Shaded stages are modified by the proposed improvements (C2: boundary-aware loss; C3: domain augmentation and test-time augmentation).

all models are scored on the same masks. Each run writes its configuration, seed, and per-image scores to a shared log. The best model is served through a Gradio [16] demo that overlays the predicted mask on an uploaded image.

5 Loss Functions and Proposed Improvements

5.1 Segmentation loss

All models are trained with the sum of binary cross-entropy and a soft Dice loss:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{BCE}}(p, y) + 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}, \tag{3}$$

where i runs over pixels and ϵ avoids division by zero. Cross-entropy gives a gradient for every pixel, and the Dice term counteracts the imbalance between the small polyp region and the large background.

Table 3: Models compared in this study ("#" refers to the course reading list).

Model	#	Key idea	Problem addressed	Priority
U-Net [1]	–	Encoder–decoder with skip connections	Reference baseline	Required
MultiResUNet [6]	59	MultiRes blocks; Res paths	Polyps of varying size; encoder–decoder semantic gap	Required
CE-Net [7]	60	DAC and RMP blocks at the bottleneck	Spatial detail lost by repeated pooling	Required
TransUNet [2]	57	CNN features, ViT encoder, U-Net decoder	Limited receptive field of CNNs	Required*
Swin-Unet [3]	56	U-shaped network of Swin Transformer blocks	Global context at lower cost than ViT	Optional*
nnU-Net [8]	58	Self-configuring U-Net pipeline	Strong baseline without manual tuning	Optional

*At least one of the two Transformer models is required.

5.2 Boundary-aware loss (C2)

Dice and cross-entropy count pixels, so a prediction that covers the polyp but places its edge a few pixels off is barely penalized, whereas HD95 is sensitive to such errors. Following Kervadec et al. [17], we add a term weighted by the signed distance map ϕ_G of the ground-truth boundary, negative inside the polyp and positive outside:

$$\mathcal{L}_B = \frac{1}{|\Omega|} \sum_{q \in \Omega} \phi_G(q)\, p(q), \qquad \mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda\, \mathcal{L}_B. \tag{4}$$

Probability placed far outside the true region costs more than probability near its edge. As recommended in [17], λ starts small and is increased during training, up to [value]. An auxiliary head that predicts the boundary map is a fallback if this term makes training unstable.

5.3 Domain augmentation and test-time augmentation (C3)

We assume that a large part of the difference between centers comes from the colour, illumination, and resolution of the endoscopes. The baseline augmentation (flips, rotation, scaling, mild brightness changes) is therefore extended with stronger photometric changes: brightness and contrast, hue and saturation shifts, gamma, blur, and JPEG compression, implemented with Albumentations [18]. At inference time, predictions are averaged over horizontal and vertical flips. The effect is measured by the change in the gap Δ of Eq. (2).

Table 4: Main experimental configuration.

Component	Configuration
Training data	1,305 images (PraNet training split minus 10% for validation)
Input size	352×352 for CNNs; 224×224 for TransUNet and Swin-Unet [confirm]
Normalization	ImageNet mean and standard deviation
Augmentation	Flips, rotation, random scaling, mild colour changes; stronger photometric set for C3
Optimizer	AdamW [21], learning rate 10^{-4}, weight decay [value]
Batch size, epochs	8–16, 100 epochs, mixed precision
Loss	BCE + Dice (Eq. 3); boundary term for C2 (Eq. 4)
Checkpoint selection	Highest validation Dice; test sets never used for selection
Framework, hardware	PyTorch [22] [version]; Google Colab or Kaggle GPU [model]
Seeds	42 [and two more if GPU time allows]

5.4 Stronger pretrained encoder (C1, optional)

If the baseline comparison is complete by week 4 of the project, the encoder of the best model is replaced by a PVTv2 [19] or ConvNeXt [20] backbone pretrained on ImageNet, while the decoder keeps the Res paths of MultiResUNet. The aim is to combine global context with multi-scale detail.

6 Experimental Setup

Table 4 summarizes the configuration. For a predicted mask $\hat{y}$ and a ground-truth mask y, the overlap and sensitivity metrics are

$$\text{Dice} = \frac{2|\hat{y} \cap y|}{|\hat{y}| + |y|}, \qquad \text{IoU} = \frac{|\hat{y} \cap y|}{|\hat{y} \cup y|}, \qquad \text{Recall} = \frac{|\hat{y} \cap y|}{|y|}. \tag{5}$$

Recall is reported because a missed polyp is more harmful than a false alarm. Boundary accuracy is measured with the 95th percentile of the symmetric Hausdorff distance,

$$\text{HD95}(\hat{y}, y) = \max\Big\{ P_{95}\big\{ d(a, \partial y) : a \in \partial \hat{y} \big\},\ P_{95}\big\{ d(b, \partial \hat{y}) : b \in \partial y \big\} \Big\}, \tag{6}$$

where ∂ denotes the boundary of a mask and d the Euclidean distance in pixels at the original resolution. Since HD95 values differ between libraries [5], a single library ([name, version]) is used for all models, and an empty prediction receives the image diagonal as its HD95. All metrics are computed per image and averaged per test set (mDice, mIoU).

Two models are compared with the Wilcoxon signed-rank test [23] on paired per-image Dice scores; a difference is called significant only when $p < 0.05$. To compare all models at once, the Friedman test [24] is followed by a post-hoc test, as recommended by Demšar [25]. Means on the out-of-domain sets are reported with 95% bootstrap

Table 5: mDice on the five test sets. PraNet values are those reported in [4] and are given for reference only.

	In-domain		Out-of-domain			
Model	Kvasir	ClinicDB	ColonDB	ETIS	CVC-300	p vs. CNN
PraNet [4]	0.898	0.899	0.709	0.628	0.871	–
U-Net	–	–	–	–	–	–
MultiResUNet	–	–	–	–	–	–
CE-Net	–	–	–	–	–	–
TransUNet	–	–	–	–	–	–
Swin-Unet	–	–	–	–	–	–

Table 6: Generalization of each model. In-domain: Kvasir and ClinicDB; out-of-domain: ColonDB, ETIS, and CVC-300 (mean over the three sets, 95% CI in brackets).

Model	mDice in	mDice out	Δ	Recall out	HD95 out (px)
U-Net	–	–	–	–	–
MultiResUNet	–	–	–	–	–
CE-Net	–	–	–	–	–
TransUNet	–	–	–	–	–
Swin-Unet	–	–	–	–	–

confidence intervals. Efficiency is reported as the number of parameters, FLOPs, and frames per second at batch size 1.

7 Results and Discussion

7.1 CNN versus Transformer models (RQ1)

[Placeholder] Compare mDice of all models on the five test sets (Table 5). State whether the best Transformer is significantly better than the best CNN, and discuss the role of ImageNet pretraining.

7.2 Out-of-domain generalization (RQ2)

[Placeholder] Report the gap Δ, recall, and HD95 on the out-of-domain sets (Table 6). Identify which model degrades least and whether the ranking changes between Dice and HD95 (Figure 3).

7.3 Ablation of the proposed improvements (RQ3)

[Placeholder] Starting from the best baseline after week 4, add each component and report its effect (Table 7). Report an improvement that is not significant as a negative result.

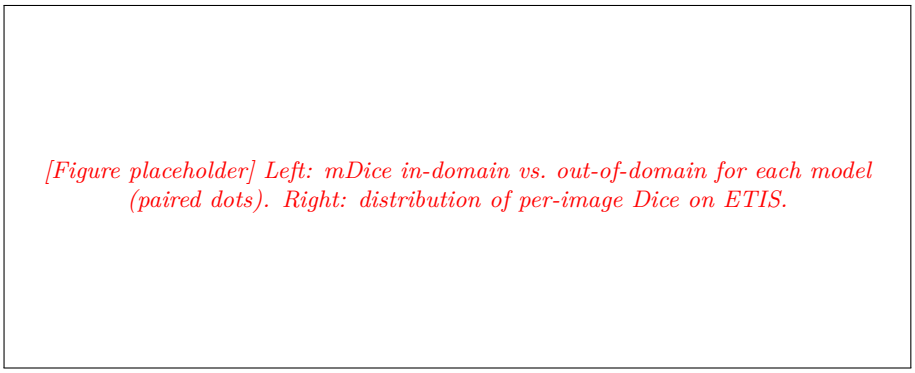


Fig. 3: Generalization gap of each model.

Table 7: Ablation on the best baseline model. p-values compare each row with the baseline on per-image out-of-domain Dice.

Configuration	mDice in	mDice out	Δ	HD95 out (px)	p
Best baseline [name]	–	–	–	–	–
+ C3 (domain aug. + TTA)	–	–	–	–	–
+ C2 (boundary loss)	–	–	–	–	–
+ C2 + C3	–	–	–	–	–
C1 + C2 + C3 (optional)	–	–	–	–	–

7.4 Error analysis and efficiency

[Placeholder] Show the lowest-Dice images of each test set (Figure 4), group results by polyp size, and compare model cost (Table 8).

8 Reliability and Limitations

The evidence is constrained in several ways. The models are trained on 1,305 images from two sources, and the out-of-domain sets are small (60–380 images), so their means carry wide confidence intervals; differences smaller than these intervals are not interpreted. All test images contain at least one polyp, so false alarms on normal frames are not measured, and still images selected by dataset creators are cleaner than live video, which contains blur, specular highlights, and motion.

Free GPU time limits the number of runs, so some configurations are trained with a single seed. Hyperparameters are shared rather than tuned per model, which may favour models whose original recipe resembles the shared one. The reproduced models are trained on different data and input sizes than in their original papers, so the scores are compared only within this pipeline and with PraNet, not with the numbers reported in the original papers. The study is an academic comparison on public data; it involves no clinical validation, and the demo is not a medical device.

Table 8: Model size and inference speed (batch size 1, [GPU]).

Model	Parameters (M)	FLOPs (G)	Frames per second
U-Net	–	–	–
MultiResUNet	–	–	–
CE-Net	–	–	–
TransUNet	–	–	–
Swin-Unet	–	–	–
Proposed	–	–	–

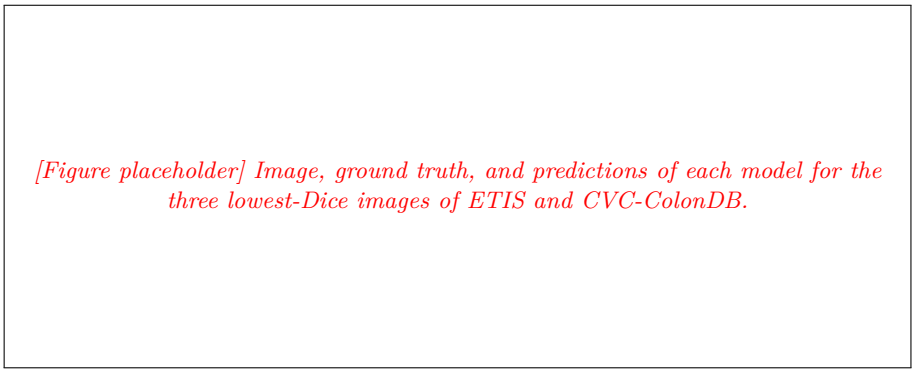


Fig. 4: Typical failure cases on out-of-domain images.

9 Conclusion and Future Work

[Placeholder] Summarize the answers to RQ1–RQ3 in one paragraph, with the main numbers.

Future work should test the models on PolypGen [9], whose six centers give a stronger external evaluation than the public benchmark sets. Evaluating on colonoscopy videos would measure stability across frames and false alarms on frames without polyps. Lighter models suited to real-time inference, and semi-supervised training with unlabelled colonoscopy frames, are further extensions.

Data and Code Availability. The project code, configurations, data split lists, per-image scores, and plotting scripts are maintained in the project repository [URL]. The datasets are public and remain subject to their original terms; they are not redistributed.

Appendix A Model-Specific Settings

Table A1: Settings that differ between models (IN: ImageNet). All other settings follow Table 4.

Model	Input size	Pretrained weights	Batch	Epochs	Params (M)
U-Net	352×352	none	–	–	–
MultiResUNet	352×352	none	–	–	–
CE-Net	352×352	ResNet-34 (IN-1k)	–	–	–
TransUNet	224×224	R50-ViT-B/16 (IN-21k)	–	–	–
Swin-Unet	224×224	Swin-T (IN-1k)	–	–	–
Proposed	–	–	–	–	–

Appendix B Reproducibility Artifact Map

Table B2: Primary artifacts in the project repository (planned layout; update to the final paths).

Purpose	Artifact
Run instructions	README.md
Data split lists	data/splits/{train,val,test}_*.txt
Duplicate check	scripts/check_duplicates.py
Experiment configurations	configs/*.yaml
Model implementations	src/models/
Losses (BCE + Dice, boundary)	src/losses.py
Training and evaluation	scripts/train.py, scripts/evaluate.py
Per-image scores	outputs/metrics/per_image_*.csv
Statistical tests	scripts/stats_tests.py
Report figures	scripts/plot_report_figures.py → report/figures/
Demo	app/gradio_app.py

References

[1] Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional Networks for Biomedical Image Segmentation. In: Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. LNCS, vol. 9351, pp. 234–241. Springer (2015)

[2] Chen, J., Lu, Y., Yu, Q., et al.: TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation. arXiv preprint arXiv:2102.04306 (2021)

[3] Cao, H., Wang, Y., Chen, J., et al.: Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation. In: Computer Vision – ECCV 2022 Workshops (2023). arXiv:2105.05537

[4] Fan, D.-P., Ji, G.-P., Zhou, T., Chen, G., Fu, H., Shen, J., Shao, L.: PraNet: Parallel Reverse Attention Network for Polyp Segmentation. In: Medical Image Computing and Computer-Assisted Intervention – MICCAI 2020 (2020). arXiv:2006.11392. Code and data split: https://github.com/DengPingFan/PraNet

[5] Urooj, A., Abdien, Z.U., Madan, N.: Metrics or Mirage? An Audit of Evaluation Inconsistencies in Colonoscopy Polyp Segmentation Benchmarks. arXiv preprint arXiv:2607.08203 (2026)

[6] Ibtehaz, N., Rahman, M.S.: MultiResUNet: Rethinking the U-Net Architecture for Multimodal Biomedical Image Segmentation. Neural Networks 121, 74–87 (2020)

[7] Gu, Z., Cheng, J., Fu, H., et al.: CE-Net: Context Encoder Network for 2D Medical Image Segmentation. IEEE Transactions on Medical Imaging 38(10), 2281–2292 (2019)

[8] Isensee, F., Petersen, J., Klein, A., et al.: nnU-Net: Self-adapting Framework for U-Net-Based Medical Image Segmentation. arXiv preprint arXiv:1809.10486 (2018). Extended version: Nature Methods 18, 203–211 (2021)

[9] Ali, S., Jha, D., Ghatwary, N., et al.: A Multi-centre Polyp Detection and Segmentation Dataset for Generalisability Assessment. Scientific Data 10, 75 (2023)

[10] Maier-Hein, L., Reinke, A., Godau, P., et al.: Metrics Reloaded: Recommendations for Image Analysis Validation. Nature Methods 21(2), 195–212 (2024)

[11] Jha, D., Smedsrud, P.H., Riegler, M.A., Halvorsen, P., de Lange, T., Johansen, D., Johansen, H.D.: Kvasir-SEG: A Segmented Polyp Dataset. In: MultiMedia Modeling (MMM 2020). LNCS, vol. 11962, pp. 451–462. Springer (2020)

[12] Bernal, J., Sánchez, F.J., Fernández-Esparrach, G., Gil, D., Rodríguez, C., Vilariño, F.: WM-DOVA Maps for Accurate Polyp Highlighting in Colonoscopy: Validation vs. Saliency Maps from Physicians. Computerized Medical Imaging and Graphics 43, 99–111 (2015)

[13] Bernal, J., Sánchez, J., Vilariño, F.: Towards Automatic Polyp Detection with a Polyp Appearance Model. Pattern Recognition 45(9), 3166–3182 (2012)

[14] Silva, J., Histace, A., Romain, O., Dray, X., Granado, B.: Toward Embedded Detection of Polyps in WCE Images for Early Diagnosis of Colorectal Cancer. International Journal of Computer Assisted Radiology and Surgery 9(2), 283–293 (2014)

[15] Vázquez, D., Bernal, J., Sánchez, F.J., et al.: A Benchmark for Endoluminal Scene Segmentation of Colonoscopy Images. Journal of Healthcare Engineering 2017, 4037190 (2017)

[16] Abid, A., Abdalla, A., Abid, A., Khan, D., Alfozan, A., Zou, J.: Gradio: Hassle-Free Sharing and Testing of ML Models in the Wild. arXiv preprint arXiv:1906.02569 (2019)

[17] Kervadec, H., Bouchtiba, J., Desrosiers, C., Granger, E., Dolz, J., Ben Ayed, I.: Boundary Loss for Highly Unbalanced Segmentation. Medical Image Analysis 67, 101851 (2021)

[18] Buslaev, A., Iglovikov, V.I., Khvedchenya, E., Parinov, A., Druzhinin, M., Kalinin, A.A.: Albumentations: Fast and Flexible Image Augmentations. Information 11(2), 125 (2020)

[19] Wang, W., Xie, E., Li, X., et al.: PVT v2: Improved Baselines with Pyramid Vision Transformer. Computational Visual Media 8(3), 415–424 (2022)

[20] Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., Xie, S.: A ConvNet for the 2020s. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 11976–11986 (2022)

[21] Loshchilov, I., Hutter, F.: Decoupled Weight Decay Regularization. In: International Conference on Learning Representations (ICLR) (2019)

[22] Paszke, A., Gross, S., Massa, F., et al.: PyTorch: An Imperative Style, High-Performance Deep Learning Library. In: Advances in Neural Information Processing Systems 32 (2019)

[23] Wilcoxon, F.: Individual Comparisons by Ranking Methods. Biometrics Bulletin 1(6), 80–83 (1945)

[24] Friedman, M.: The Use of Ranks to Avoid the Assumption of Normality Implicit in the Analysis of Variance. Journal of the American Statistical Association 32(200), 675–701 (1937)

[25] Demšar, J.: Statistical Comparisons of Classifiers over Multiple Data Sets. Journal of Machine Learning Research 7, 1–30 (2006)